Moving Auctions Online

Yunmi Kong[†] Shosuke Noguchi[‡]

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Abstract

This paper documents the impact of a simple and universally applicable auction innovation: moving online. We leverage bid-level data from an offline-to-online transition of state mineral lease auctions, the features of which facilitate a detailed study. Controlling for auction heterogeneity, we find a large revenue difference in favor of online auctions, explained by a large increase in the number of bidders. Examining bidders' identities reveals an inflow of new participants who had not participated before the transition. We do not detect different post-auction outcomes between new and existing participants. Where not yet adopted, online auctions are well worth consideration.

Keywords: auction, competition, government, online, internet, digital

We would like to thank the New Mexico State Land Office for providing the data. All remaining errors are ours.

[†]Department of Economics, Rice University.

[‡]School of Commerce, Waseda University.

E-mail: yunmi.kong.01@gmail.com, shosuke.noguchi@waseda.jp

1 Introduction

Governments and public entities around the world buy and sell a vast array of goods and services through auctions. Well-known examples from the economics literature include the procurement of construction services and the sale of timber, oil and gas leases, treasury bonds, and radio spectrum rights, among many others. As surveyed by Hortaçsu and Perrigne (2021), thirty-plus years of empirical analysis have been dedicated to the study and improvement of these auctions, in the hope of ultimately benefiting the public.

In that spirit, this paper aims to document the impact of a simple and universally applicable auction innovation: moving online. While auctions previously involved bidding by paper mail or participating in person at a live auction, auctions today can be implemented fully on the internet, such as through an online bidding platform. Interestingly, some auction programs still (at the time of writing) run in-person oral auctions or receive sealed bids at a designated physical address; federal timber auctions in the U.S. are a major example. In some other cases, the transition to online auctions is surprisingly recent; for example, the Bureau of Land Management (BLM)'s oil and gas leasing program announced its move to online auctions in August 2016.[1] This is because the move first required an amendment by Congress of the Mineral Leasing Act, which previously authorized oral auctions only. While oil and gas leasing programs on individual states' trust lands are not under the authority of the Mineral Leasing Act, major programs such as that of New Mexico made permanent moves online only around or after the BLM announcement. These recent transitions offer good opportunities for empirical analysis, because reliable auction records before and after the transition are more likely to be accessible. Our goal is to provide practitioners with detailed and direct evidence of the potential benefits of online auctions.

To document an offline to online transition in detail, we leverage data from oil and gas leases auctioned by the New Mexico State Land Office (NMSLO), which offer a number of advantages for studying this topic. Their auctions have been a recurring subject of study in the economics literature[2] due to the relatively high volume of auctions, detailed bidding data, and their location in the Permian Basin, one of the most prolific oil and gas basins in the world. Moreover, they are an important source

[1] www.blm.gov/press-release/blm-takes-important-step-toward-online-oil-gas-lease-sales

[2] See, for example, Kong (2020), Kong (2021), Bhattacharya, Ordin and Roberts (2022), Ordin (2019), and Martin, Pesendorfer and Shannon (2025).

of funds for the state; the state trust lands managed by the NMSLO supply about one quarter of the budgets for all the public school districts in New Mexico.[3] Beyond these features, the NMSLO auctions offer three key advantages for evaluating the offline to online transition as directly as possible: i) observability of all bids and bidder names via sealed bids, ii) one-dimensional bids, and iii) detailed data on key determinants of item value, such as oil and gas prices and tract-level production.

To elaborate, the first advantage is that they use sealed-bid auctions (in addition to ascending auctions), both before and after the transition. Ascending auctions, also known as oral, open outcry, or English auctions, are a common auction format for offline government auctions, but the associated records typically lack information on losing bids and losing bidders due to the nature of live, in-person bidding. In contrast, sealed-bid auctions receive formal written bids that are recorded, allowing us to compare not only final auction prices but also all the submitted bids and number and identity of bidders before and after moving online. We would not be able to get under the hood of what is happening, to the extent that we do in this paper, with a dataset of only ascending auctions. The second advantage of the NMSLO auctions is that the bids are one-dimensional, consisting of a scalar dollar amount, allowing a direct comparison of bids and auction results. Some government auctions, such as those for construction procurement, feature multidimensional bidding, in which the bid consists of a vector of components. With multidimensional bids, the bid space is not intrinsically ordered, so evaluation of what is better or worse depends on an additional layer of judgment. The third advantage is that data on key drivers of tract value, such as oil and gas prices, tract-level production, and historical value by geographical location, allow for tight control of confounding factors.

The NMSLO began a transition from offline to online auctions in September 2016, immediately following the BLM's August announcement of its own plans to move online. The NMSLO used online auctions exclusively from January 2017 onwards. We document before-and-after differences in deflated auction prices, submitted bids, and, importantly, the number and identity of bidders using auctions from October 2015 to December 2018. We choose this window because the NMSLO employed a stable reserve price policy during this time; the reserve price policy differed before and shortly after this time period, making bids less comparable outside of this window. We take care to control for terms of the lease contract, variables that change with time

[3] www.nmstatelands.org/about/trust-beneficiaries

(such as oil and gas prices and operating cost indices), and tract-level heterogeneity by leveraging historical auction and production data in our regression specifications.

According to our regression estimates, online auctions are associated with significantly higher sales prices (i.e., winning bids) compared to offline auctions; the price per acre increases by 79% on average. In sealed-bid auctions, for which we observe losing bids as well as winning bids, the winning bid per acre more than doubles, while the submitted bid per acre increases by one third. These large positive relationships are corroborated by supplementary analysis of subsamples and of alternative control strategies. To explain this revenue increase, we examine the level of competition as represented by the number of bidders. Conditional on auction covariates, we find that online sealed-bid auctions are associated with about two more bidders per auction on average compared to offline sealed-bid auctions, so that the average number of bidders changes from about three to five. This is a very meaningful increase in competition. Conditional on having the same number of bidders, bids submitted to online versus offline sealed-bid auctions do not exhibit a significant difference, confirming increased competition as the primary driver of the online auction effect.

Next, we exploit the availability of bidder names in the data to examine the source of this increased competition and, in particular, the degree to which moving online drew in new participants. We define a "new" bidder as a bidder name that appears in an online auction but does not appear in the preceding decade of offline auction data. The average number of new bidders per auction in the online sealed-bid auctions is about two, which roughly corresponds to the total online-offline difference in the number of bidders. Indeed, we find no statistically significant change in the number of *existing* (not new) bidders as the auctions go from offline to online, suggesting that the documented increase in competition is driven primarily by an inflow of new bidders. Finally, we investigate the potential concern that post-auction outcomes, such as the rate of drilling and production, may differ between leases won by new versus existing bidders. While the evidence should be interpreted with caution because drilling is fairly rare across the board, we find no evidence in our data that the probability of drilling or volume of production is different between these two groups.

Our findings suggest that, for those auction programs that are currently run offline, online auctions would be well worth a trial. Of course, every auction program is different, and it is possible that the NMSLO constitutes a best-case scenario in terms of revenue improvement. Nonetheless, the evidenced potential for increased competition is important. Bulow and Klemperer (1996) prove that a simple auction with $N+1$ bidders yields the seller higher expected revenue than she could earn even by fully exploiting her monopoly position against N bidders. In particular, they remark that,

> Since the informational demands for computing optimal mechanisms are substantial, and the computations involved are complex, this result suggests that it will often be more worthwhile for a seller to devote resources to expanding the market than to collecting the information and making the calculations required to figure out the best mechanism.

Indeed, the simplicity of the innovation relative to the magnitude of potential market expansion suggests that moving auctions online deserves priority among design interventions for government auctions.

Corroborative evidence from outside of New Mexico provides external validity regarding the revenue effect of moving online. Karsten (2020) studies oil and gas lease auctions run by the North Dakota Department of Trust Lands, which alternated between offline and online ascending auctions during 2012–2016 before transitioning permanently to online auctions in August 2016.[4] During the period of alternation, he found that online auctions brought in between 20% and 160% higher auction prices per acre than an offline auction for a similar lease. The number of bidders in the offline North Dakota auctions is not observed, so Karsten (2020) could not directly assess changes in competition. Using a calibrated auction model, he proposed that a difference of about four bidders per auction would rationalize the observed revenue difference. When it comes to federal auctions, the amendment of the Mineral Leasing Act discussed above was preceded by a successful online auction pilot conducted by the BLM in Colorado in 2009, which led to internal projections of higher revenue upon moving online.[5] The BLM, like the North Dakota Department of Trust Lands, used ascending oral auctions prior to their online transition, so their auction data do not allow an assessment of competition either. In this paper, we now document direct evidence that online auctions can attract new participants that substantially increase

[4]When we combine the New Mexico and North Dakota data and run a two-way (state and time) fixed effects regression, we estimate an even larger revenue effect of online auctions than what we present in this paper. However, there is external evidence that the oil and gas industries of different states generally do not experience parallel trends, so we do not present that analysis here.

[5]www.blm.gov/press-release/blm-announces-additional-online-oil-and-gas-lease-sales-2016

competition. On the related but distinct question of using centralized auctions versus informal negotations to sell leases, Covert and Sweeney (2023) find that auctions yield over 40% higher revenue in Texas.

In some developing countries, public procurement, especially through manual processes, has a reputation for auctioneer corruption involving bribery, blocking access to necessary bid-preparation documents, or even physical intimidation. In these contexts, electronic procurement processes that make all procurement data public can improve outcomes by increasing transparency. Lewis-Faupel et al. (2016) find that e-procurement led to project quality improvements in India and both quality and delay improvements in Indonesia. In Ukraine, Kovalchuk, Kenny and Snyder (2019) discuss the impact of e-procurement, while Deininger, Ali and Neyter (2023) find that transparent online auctions increased lease revenue from public agricultural land. In Thailand, Santoso and Bourpanus (2019) survey contractors of public projects and find that fully electronic procurement changed the factors that influence bidders' markups. Meanwhile, the evidence in our paper shows that, even in environments where auctioneer corruption is not a concern, moving auctions online can bring large improvements that benefit the taxpayer.

More broadly, the management, marketing, and supply chain literatures have studied the role of online auctions in business-to-business and business-to-consumer transactions, as surveyed by Jap (2002), Pinker, Seidmann and Vakrat (2003), and Mora Cortez, Cabanelas and Charterina (2023). For example, in the context of wholesale used car auctions, Overby and Jap (2009) explain how buyers and sellers used electronic and physical channels when both were simultaneously available, and Overby and Forman (2015) study how the availability of an electronic channel affected geographic trading patterns and price dispersion through increased remote purchasing.

As discussed earlier, the New Mexico State Land Office auctions have been a recurring subject of study in the economics literature. The rich set of topics and auction models studied thereby include sealed-bid versus ascending auctions (Kong (2020)); dynamics across sequential auctions (Kong (2021)); bidding in light of endogenous drilling (Bhattacharya, Ordin and Roberts (2022)); distortionary effects of taxes (Ordin (2019)); and tests of Bayesian Nash equilibrium and underbidding (Martin, Pesendorfer and Shannon (2025)). In this paper, we do not specify a bidding model. We expect the qualitative benefits of online auctions—stronger competition leading to improved auction revenue—to be generally applicable across a wide variety of bidding models and auction programs of various goods and services.

The NMSLO papers above are part of a broader literature that derives economic insights from oil and gas lease auctions, going back to Hendricks, Porter and Boudreau (1987) and Hendricks and Porter (1988)'s analysis of the U.S. Outer Continental Shelf. We refer readers to Hortaçsu and Perrigne (2021), Chapter 4 for a thorough survey.[6]

2 Background and Data

This section introduces the auction institution, presents preliminary patterns in the raw data, and describes the covariates that characterize each auction item.

2.1 Background

As discussed in the introduction, public entities use auctions extensively both to buy (procure) and to sell. State trust lands in the United States are a major example that use auctions to raise funds. These lands were granted by Congress to states, primarily in the Western U.S., upon their entrance into the Union for the purpose of funding public education through land leasing or the sale of renewable resources. According to the National Association of State Trust Lands, their "combined holdings [...] represent the second largest owner of land and minerals in the United States, second only to the federal government."[7]

The New Mexico State Land Office (NMSLO) runs a particularly active auction program to lease its oil-and-gas-rich lands in the Permian Basin. According to the NMSLO, about 94 percent of their earnings go to support public education, and they supply about one quarter of the budgets for all the public school districts in New Mexico.[8] Bids in these auctions are one-dimensional, consisting of scalar dollar amounts for the lease "bonus," which refers to the cash that the winning lessee pays the lessor at the start of the lease. Historically, the NMSLO used both first-price sealed-bid and ascending auctions concurrently. Bidding occurred by mail for the first-price sealed-bid auctions, while the ascending auctions were run live, in person.

[6] See, for example, Hendricks, Porter and Spady (1989); Hendricks and Porter (1992, 1993, 1996); Hendricks, Porter and Tan (1993, 2008); Hendricks, Porter and Wilson (1994); Porter (1995); Hendricks, Pinkse and Porter (2003); Haile, Hendricks and Porter (2010); Aradillas-Lopez et al. (2018); Sant'Anna (2018); Compiani, Haile and Sant'Anna (2020); and Kong, Perrigne and Vuong (2022).

[7] www.statetrustland.org

[8] www.nmstatelands.org/about/trust-beneficiaries

Then, in the last quarter of 2016, the NMSLO transitioned to online auctions, so that bidders now bid through an internet bidding platform. Our goal is to document the impact, if any, of transitioning from offline to online auctions.

We define our data sample of NMSLO auctions as follows. First, the time period we study, surrounding the transition to online auctions, is from October 2015 through December 2018. This is a time period during which the NMSLO's reserve price policy was stable; it assigned the minimum acceptable bid (reserve price) for a given tract as a weakly increasing step function of its acreage. The NMSLO used a different reserve price function prior to October 2015, and it changed its reserve price policy again in early 2019. Second, the geographic area we study is the Permian Basin. In our sample, we define this to include tracts in the Chaves, De Baca, Eddy, Lea, and Roosevelt counties of New Mexico. These comprise 99% of all auctioned tracts during this period. Finally, all lease contracts include a royalty rate, which the lessee must pay the NMSLO if there is oil or gas production on the leased tract. For Permian Basin tracts in this time period, 98% of tracts had a royalty rate of either 0.1875 or 0.20—we discuss this in more detail later—but 2% of tracts had a lower rate. We exclude this 2% from our sample. Then, to summarize, our analysis sample consists of Permin Basin leases sold via NMSLO auctions during October 2015–December 2018, with a royalty rate of either 0.1875 or 0.20.

2.2 A transition to online auctions

The New Mexico State Land Office (NMSLO) holds one auction date per month, with multiple leases being auctioned on each date. Before the transition, NMSLO auctions were exclusively offline, with 102 leases sold by sealed-bid auctions and 90 leases sold by ascending auctions during the eleven months preceding the transition. Then, the first online auctions took place in September 2016, beginning a transitional period lasting through December 2016 in which both offline and online auctions were used. Specifically, there were 54 online ascending auctions, 34 offline ascending auctions, and 9 offline sealed-bid auctions during the transitional period. Finally, the auctions were exclusively online from January 2017 onwards. During the two years hence, 361 leases were sold by online sealed-bid auctions, and 190 leases were sold by online ascending auctions.

To gain a preliminary understanding of how auction revenue evolved during our

Figure 1: Sealed-bid auction revenue by auction date

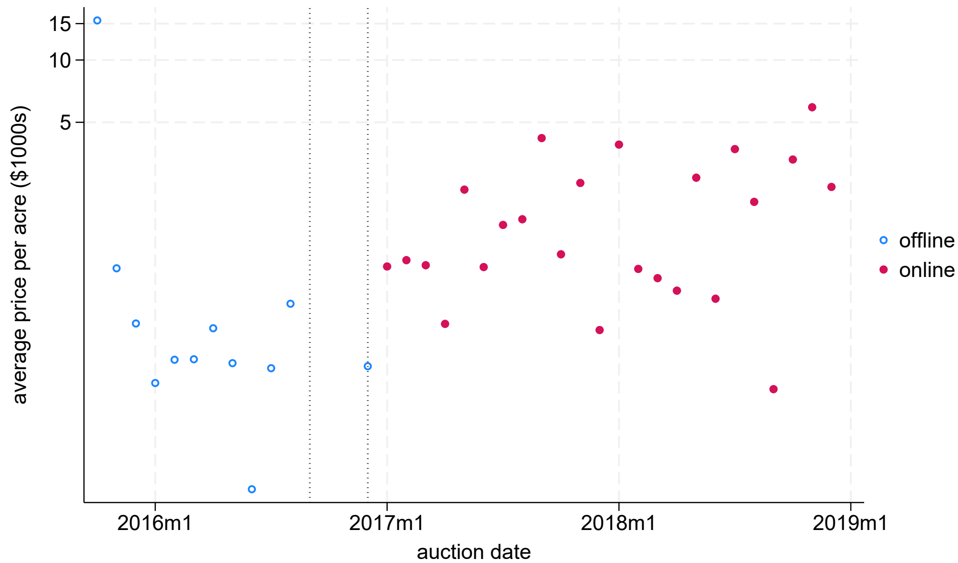
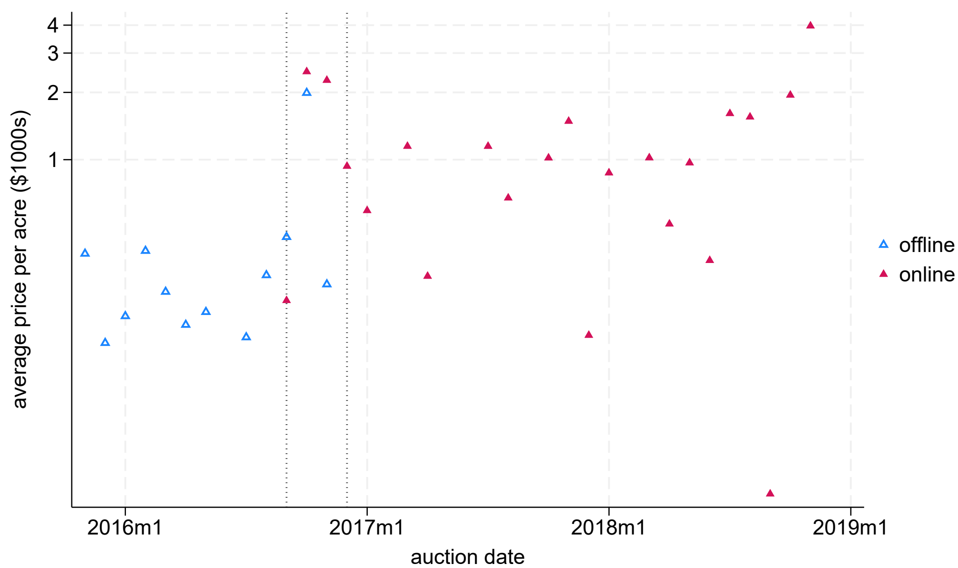


Figure 2: Ascending auction revenue by auction date



Notes: Each marker includes auctions from one auction date; auctions occur once a month. The average price per acre for an auction date and auction format is computed as the total relevant auction revenue divided by total auctioned acreage. The y-axes are on a log scale and units are in thousands of 2015 dollars. The dotted vertical lines mark the transitional period of September–December 2016 during which the NMSLO used both offline and online auctions.
The unusually high marker for the October 2015 sealed-bid auction comes from only a single lease on that date satisfying our sample criteria (see the end of Section 2.1), which happened to sell at a very high price.

sample period, Figures 1 and 2 plot the average price per acre the NMSLO received on each auction date, for sealed-bid auctions and ascending auctions, respectively. To allow comparison of prices over time, all dollar amounts have been deflated to 2015 dollars using the GDP deflator for the U.S.[9] Each marker includes auctions from one auction date, and the dotted vertical lines indicate the beginning and end of the transitional period of September–December 2016. Unfilled markers represent offline auctions, while filled markers represent online auctions. Note the y-axes are on a log scale.

Visually, Figures 1 and 2 suggest an increase in auction revenue that coincides with the adoption of online auctions. While these raw statistics are intriguing, a careful evaluation requires consideration of a number of covariates, or variables that describe the heterogeneity of different tracts and of different dates. We describe these next.

2.3 Covariates

The variables describing an auctioned lease broadly fall into three categories: i) contract terms, ii) variables that vary with time, and iii) variables that vary with tract location. When it comes to the first category, some contract terms—such as the lease duration, which is five years—do not vary across leases in our sample. The key terms that do vary are an annual rental payment called the "primary rental" and the royalty rate. The primary rental is determined by the location of the tract and takes one of two possible values: fifty cents per acre or one dollar per acre. We capture this variation with a dummy variable. Meanwhile, royalty rates in our sample do not vary cross-sectionally but only across time: all leases in our sample have a royalty rate of 0.1875 through January 2017; subsequently, the NMSLO began exclusive use of 0.20 as the royalty rate from February 2017 through December 2018. Note that higher royalty rates are *less* attractive to bidders, who must pay that portion of any production revenue to the NMSLO. In the regression analysis that follows in Section 3, we do not include an indicator for 0.20 versus 0.1875 royalty rates because such an indicator would be largely collinear with an online auction indicator. As the higher rate adopted later is less attractive to bidders, this omission is likely to negatively bias the estimated effect of online auctions on auction prices, going against finding

[9] The GDP deflator is published by the Organization for Economic Co-operation and Development (OECD).

the positive effect suggested in Figures 1 and 2.

The second category, variables that vary with time, includes indices of the revenues and costs of oil and gas exploration and production. In particular, industry trends are strongly driven by oil and gas prices. We use the West Texas Intermediate spot crude oil price published by the Federal Reserve Bank of St. Louis (FRED) and the Henry Hub natural gas spot price published by the U.S. Energy Information Administration (EIA) to account for variation in revenue per produced unit. To account for variation in industry costs, we include the Upstream Capital Cost Index (UCCI) and the Upstream Operating Costs Index (UOCI) published by S&P Global Commodity Insights. As we deflate auction prices using the GDP deflator to eliminate inflation, we apply the same deflation to these commodity prices and cost indices as well.

The third category of variables includes the geological information implied by a tract's location on the Public Land Survey System (PLSS) grid. First, we combine the tract location descriptions in the auction data with well data obtained from Drilling-info (Enverus) to map oil and gas wells to auctioned tracts. From this mapping, we derive the drilling and production history of a tract: whether it has been drilled and the barrel-of-oil-equivalents (BOE) produced in that history. This is relevant because firms' revenues from leasing a tract are driven by expected production. For each lease, we obtain both the pre-auction production history and the post-auction production outcome, i.e. the cumulative BOE produced from wells spudded after the auction. Second, we exploit twenty years of historical auction data and the fact that geological attractiveness is continuous in space[10] to more finely control for unobserved tract heterogeneity as follows. Following Kong (2020) and Kong (2021), we construct a "heatmap" index of location-based heterogeneity. At a high level, this index represents the predicted attractiveness of a tract based on a nonparametric surface that fits historical bids to geographic coordinates. The idea is that the index should be high (low) for tracts near areas that have historically attracted high (low) bids. The historical bids are from sealed-bid auctions run by the NMSLO during 1995-2014, so they are external to our main data sample. We deflate these historical bids using the GDP deflator and then fit a smooth surface of the log bids per acre on geographic (north-south and east-west) coordinates using local quadratic regression. The fitted

[10]Nearby tracts share similar geology; Hodgson (2018) explains that industry techniques for predicting geological features use spatial interpolation.

Table 1: Summary statistics of covariates by auction format

	offline		online	
	sealed	ascending	sealed	ascending
heatmap index	4.59	4.51	4.75	4.64
	(0.41)	(0.38)	(0.43)	(0.39)
drilled before	0.44	0.48	0.50	0.43
	(0.50)	(0.50)	(0.50)	(0.50)
cumulative BOE pre-auction (log)	3.75	3.56	3.87	2.85
	(5.37)	(5.22)	(5.39)	(4.91)
cumulative BOE post-auction (log)	0.67	0.00	0.41	0.60
	(2.78)	(0.00)	(2.23)	(2.65)
primary rental=$1/acre	0.51	0.42	0.57	0.45
	(0.50)	(0.50)	(0.50)	(0.50)
acres (in units of 100)	2.61	2.38	2.43	2.49
	(0.93)	(1.06)	(1.00)	(0.98)
WTI spot crude oil price (log)	3.71	3.73	3.98	3.94
	(0.15)	(0.14)	(0.14)	(0.14)
Henry Hub Natural Gas Spot Price (log)	0.83	0.84	1.11	1.07
	(0.21)	(0.19)	(0.13)	(0.09)
Upstream Capital Cost Index	169.66	169.03	171.15	169.93
	(3.08)	(2.67)	(1.90)	(2.18)
Upstream Operating Cost Index	166.32	165.65	165.78	165.71
	(3.44)	(2.87)	(2.04)	(1.59)
Observations	111	124	361	244

Table displays sample means of the covariates, with sample standard deviations in parentheses.

surface yields a heatmap index for every auctioned lease in our sample. Finally, we also account for the acreage of the tract. Table 1 provides summary statistics of the covariates by whether the auction is offline or online and sealed-bid or ascending.

3 Empirical Analysis

In this section, we document differences in revenue before and after the auctions move online. To explain this difference, we then investigate changes in the number and identity of bidders, of which the direct observability is a key advantage of this data. Finally, we consider effects on post-auction outcomes, namely drilling and production.

3.1 Sales prices (winning bids)

We now examine whether there is an online versus offline difference in the sales price the State Land Office was able to obtain for its auctioned leases, i.e., the winning bids, conditional on the auction covariates discussed in Section 2.3. Table 2, columns (1)–(4) regress the log winning bid per acre on an online auction indicator, including the covariates as regressors. Columns (1) and (2) use both sealed-bid and ascending auctions in the regression sample, while column (3) uses only sealed-bid auctions and column (4) uses only ascending auctions.

In Table 2, column (1), online auctions are associated with significantly higher sales prices conditional on auction covariates. An increase in log price per acre of 0.583 implies a percentage increase of $\exp(0.583) - 1 \approx 79\%$. Meanwhile, sealed-bid auctions have higher sales prices than ascending auctions, a phenomenon previously documented and analyzed in Kong (2020) using pre-2015 offline auctions. Kong (2020) proposed bidders' risk aversion combined with uncertainty about the number of competitors as an explanation for the revenue non-equivalence. Interestingly, the NMSLO eventually phased out the use of ascending auctions from August 2019 onwards. In column (2), interacting the online auction indicator with the sealed-bid auction indicator suggests that the revenue increase associated with moving online is larger for sealed-bid auctions. In general, when any of the assumptions underlying the Revenue Equivalence Theorem do not hold, auction revenue in sealed-bid and ascending auctions need not respond the same way to changes in the auction environment, such as changes in the distribution of the number of bidders. In Table 2, column (3), which analyzes sealed-bid auctions only, the sales price increase associated with moving online is a log increase of 0.850, or percentage increase of $\exp(0.850) - 1 \approx 134\%$. In column (4), the analogous estimate for ascending auctions only is a log increase of 0.294, or percentage increase of $\exp(0.294) - 1 \approx 34\%$, but this is not statistically significant at conventional levels, with a p-value of 0.130.

We conduct subsample analyses to confirm that these estimates are not driven by relatively cheap leases. The top quartile of leases in terms of heatmap index contribute about 70% of total auction revenue, so the impact of moving online on

Table 2: Regression analysis of sales prices and bids

	(1)	(2)	(3)	(4)	(5)
	log winning bid/acre				log bid/acre
online	0.583	0.407	0.850	0.294	0.287
	(0.149)	(0.165)	(0.220)	(0.194)	(0.114)
sealed bid auction	0.613	0.352			
	(0.083)	(0.154)			
online × sealed bid auction		0.372			
		(0.179)			
heatmap index	1.544	1.542	1.432	1.614	0.921
	(0.157)	(0.157)	(0.195)	(0.274)	(0.130)
drilled before	0.046	0.036	0.011	0.022	-0.005
	(0.102)	(0.102)	(0.152)	(0.141)	(0.100)
cumulative BOE pre-auction (log)	0.015	0.015	0.029	-0.000	0.033
	(0.011)	(0.011)	(0.015)	(0.015)	(0.010)
cumulative BOE post-auction (log)	0.099	0.102	0.072	0.145	0.025
	(0.016)	(0.016)	(0.021)	(0.021)	(0.016)
primary rental=$1/acre	0.084	0.088	0.185	-0.016	0.068
	(0.118)	(0.118)	(0.152)	(0.195)	(0.094)
acres (in units of 100)	-0.000	0.004	0.034	-0.033	-0.063
	(0.044)	(0.044)	(0.060)	(0.063)	(0.039)
WTI spot crude oil price (log)	0.606	0.598	0.003	1.701	-0.052
	(0.393)	(0.390)	(0.480)	(0.789)	(0.303)
Henry Hub Natural Gas Spot Price (log)	-0.283	-0.302	-0.206	-0.654	-0.286
	(0.339)	(0.338)	(0.465)	(0.533)	(0.284)
Upstream Capital Cost Index	-0.016	-0.021	0.024	-0.108	0.028
	(0.031)	(0.031)	(0.039)	(0.069)	(0.023)
Upstream Operating Cost Index	-0.018	-0.014	-0.013	0.019	-0.035
	(0.029)	(0.029)	(0.035)	(0.064)	(0.021)
constant	1.727	1.933	-3.050	7.133	2.361
	(3.625)	(3.663)	(5.240)	(5.061)	(3.134)
Sample: sealed (S) or ascending (O) auctions	S+O	S+O	S	O	S
Observations: auction-level (a) or bid-level (b)	a	a	a	a	b
Observations	840	840	472	368	2002
R^2	0.417	0.420	0.377	0.376	0.123
Adjusted R^2	0.409	0.411	0.362	0.357	0.118

Heteroskedasticity-robust standard errors are provided in parentheses.

the best leases is especially important. We repeat the Table 2, column (1) regression separately for “top” leases (in the top quartile of heatmap indices) and for non-top leases. The estimated sales price increase associated with moving online is a log increase of 1.359 for top leases and 0.434 for non-top leases, respectively, indicating even larger revenue increases for the best leases. Appendix Table A1, columns (1)-(2) provide the subsample regression estimates in full.

So far, we have discussed changes in sales prices, or winning bids. In Table 2, column (5), we additionally run a bid-level regression of all sealed bids, regardless of whether they won or lost. Note that a bid-level comparison of offline and online auctions is feasible only for sealed-bid auctions; offline ascending auctions were conducted orally, and bids other than the final sales price were not recorded. We find that online auctions are associated with higher bids at the bid level as well, with a log difference of 0.287.

Finally, to check the robustness of the *online* coefficents in Table 2 to more flexible control strategies, we employ the post-double-selection method of Belloni, Chernozhukov and Hansen (2014) to select control variables from the full menu of covariates, their second-order terms, and interactions while avoiding overfitting. The results, provided in Appendix Table A2, columns (1)-(5), are consistent with Table 2. We explore the source of these relationships in more depth below.

3.2 Competition

3.2.1 Number of bidders

As a potential source of the revenue increase documented in Section 3.1, we examine the level of competition as represented by the number of bidders. Note that there is no record of the number of bidders for offline ascending auctions, which were conducted orally. Therefore, we focus on sealed-bid auctions for our analysis of competition. The raw average number of bids submitted per auction is 3.08 in the offline sealed-bid auctions and 4.60 in the online sealed-bid auctions. Table 3, column (1) examines whether there is an online versus offline difference in the number of bidders per auction after controlling for the auction covariates discussed in Section 2.3. We find that online sealed-bid auctions are associated with nearly two more bidders per auction on average. Given that the average number of bidders in the offline sealed-bid auctions was about three, this is a relatively large increase in competition. Bulow

Table 3: Regression analysis of the number of bidders in sealed-bid auctions

	(1) number of bidders	(2) log winning bid/acre	(3) log bid/acre
online	1.968	0.182	-0.109
	(0.336)	(0.172)	(0.115)
heatmap index	1.466	0.872	0.674
	(0.332)	(0.171)	(0.130)
drilled before	-0.313	0.148	0.104
	(0.235)	(0.133)	(0.100)
cumulative BOE pre-auction (log)	0.066	0.004	0.014
	(0.023)	(0.013)	(0.010)
cumulative BOE post-auction (log)	0.103	0.054	0.004
	(0.051)	(0.016)	(0.017)
primary rental=\$1/acre	0.486	0.019	0.027
	(0.248)	(0.131)	(0.094)
acres (in units of 100)	-0.002	-0.003	-0.026
	(0.111)	(0.050)	(0.039)
WTI spot crude oil price (log)	-0.555	0.268	0.112
	(0.726)	(0.377)	(0.295)
Henry Hub Natural Gas Spot Price (log)	-0.452	-0.133	-0.077
	(0.682)	(0.402)	(0.281)
Upstream Capital Cost Index	-0.222	0.089	0.076
	(0.062)	(0.031)	(0.023)
Upstream Operating Cost Index	0.110	-0.057	-0.058
	(0.054)	(0.028)	(0.021)
constant	17.774	-6.042	-2.025
	(7.658)	(4.468)	(3.077)
Number of bidder fixed effects	–	Yes	Yes
Observations: auction-level (a) or bid-level (b)	a	a	b
Observations	472	472	2002
R^2	0.258	0.612	0.168
Adjusted R^2	0.240	0.591	0.158

Heteroskedasticity-robust standard errors are provided in parentheses.

and Klemperer (1996) prove that, under certain conditions, a simple auction with $N + 1$ bidders yields the seller higher expected revenue than she could earn even by fully exploiting her monopoly position against N bidders. The derived wisdom is that attracting an additonal bidder is likely to have more impact on revenue than other commonly considered improvements to auction design. Going from three bidders to five bidders, then, one would expect a considerable increase in auction revenue, as seems to be borne out by the results in Section 3.1.

The increase in bidders associated with online auctions is robust to more flexible control strategies for the covariates (Appendix Table A2, column (6)). Meanwhile, subsample analysis reveals that in the case of "top" leases (in the top quartile of heatmap indices), online auctions are associated with 3 more bidders, compared to 1.4 more bidders for non-top leases (Appendix Table A1, columns (3)-(4)). This is consistent with the larger log sales price increase experienced by top leases in Section 3.1.

In Table 2, columns (3) and (5), we saw that online sealed-bid auctions yield both higher winning bids and higher submitted bids than offline sealed-bid auctions. We now examine whether any such difference remains after controlling for the number of bidders. Table 3, columns (2) and (3) regress winning and all submitted sealed bids, respectively—in log dollars per acre, to be precise—on an online auction indicator, number of bidder fixed effects, and the auction covariates. We fail to detect a statistically significant online-offline difference in either winning or submitted bids when the number of bidders is the same. In other words, there is no evidence of higher bidding *conditional on the level of competition.* Thus, we can conclude that the observed revenue increase is driven primarily by an increased number of bidders as opposed to more aggressive bidding given the same number of bidders. Then, as the increase in number of bidders is an important mechanism, we investigate the source of this increase below.

3.2.2 Identity of bidders

We exploit the observability of bidder names in the data to examine the degree to which moving online drew in new participants. We observe all bidder names in sealed-bid auctions and winner names in ascending auctions. Any analysis of identities using reported bidder names must be interpreted with caution, because there is no guarantee of a one-to-one mapping between unique bidder names and unique bidder identities.

Also, we do not observe losing bidders in ascending auctions. Acknowledging these shortcomings, we focus on distinguishing "new" bidders from existing bidders, setting a high bar to qualify as a new bidder. Before identifying new bidders, we first manually clean the bidder names in our data sample to catch, as best as we can, variations of names that actually represent one entity. Among these cleaned names, we then tag any names of which the first appearance is in an online auction. In a final step, we consult the historical auctions preceding our data sample. Bidder names that share a first word with any historical bidder names within ten years of the start of our sample do not qualify as a new bidder. Thus, we define a "new" bidder as a bidder name that appears in an online auction but does not appear in the last decade of offline auction data. Within this group of new bidders, we also make note of any "renewed" bidders; these are bidder names that last appear before 2005 and appear again in the online auctions with no appearances in the decade between.

The raw average number of new bidders per auction in the online sealed-bid auctions is 2.06, which roughly corresponds to the online-offline difference in number of bidders estimated in Table 3, column (1). In Table 4, column (1), we examine this link further by regressing the number of existing (not new) bidders in sealed-bid auctions on an online auction indicator, controlling for the auction covariates discussed in Section 2.3. The table suppresses estimated coefficients on the auction covariates to focus on the variables of interest. We find that there is no statistically significant change in the number of existing bidders as the auctions go from offline to online, confirming that the increase in competition documented in Section 3.2.1 is driven primarily by an inflow of new bidders.

Are new bidders simply firms that did not exist before the online transition? To investigate, we manually gathered information on the firms that bid most frequently, though most were privately held and in some cases illusive. Of the top five new bidders in terms of sealed bids contributed,[11] only one (the fourth place firm) was newly formed after the transition. These bidders began participating either during the transitional period or the first quarter after the full transition, with the exception of the newly formed firm which appears later. While four out of the top five most frequent existing bidders were based locally in New Mexico, the top five new bidders

[11] They are International Petroleum, LLC; Straight Profit, Inc.; Conquest Land Company, LLC; Ridge Runner Resources, LLC; and Browning Oil Company, Inc. These names comprise a little over half of all sealed bids from new bidders in the sample.

Table 4: Regression analysis of existing and new bidders

	(1) # existing bidders	(2) probit: win	(3) bid rank
online	-0.019		
	(0.265)		
new bidder		0.232	0.006
		(0.073)	(0.012)
Auction covariates	yes	yes	yes
Number of bidder fixed effects	no	yes	yes
Auction format	all sealed	online sealed	online sealed
Observations: auction-level (a) or bid-level (b)	a	b	b
Observations	472	1642	1660
R^2	0.188		0.000
Adjusted R^2	0.169		-0.015
Pseudo R^2		0.059	

Notes: Column (1)–OLS regression; the dependent variable is the number of existing bidders; the data sample is all sealed-bid auctions; observations are at the auction level. Column (2)–probit regression; the dependent variable is a dummy variable for winning the auction; the data sample consists of online sealed-bid auctions; observations are at the bid level. The winner is perfectly predicted for auctions with one bidder; such auctions (18 observations) are automatically dropped. Column (3)–OLS regression; the dependent variable is normalized bid rank as defined in the text; the data sample consists of online sealed-bid auctions; observations are at the bid level. Estimated coefficients on auction covariates and number of bidder fixed effects are not displayed. Heteroskedasticity-robust standard errors are provided in parentheses.

were based in Utah, Texas, Oklahoma, Texas, and Texas, respectively.

Next, we examine the identity of online auction *winners*. We find that "new" bidders won about 45% of all online auctions in our sample. Of this 45%, "renewed" bidders make up 8 percentage points. To study whether new bidders are more or less likely to win conditional on submitting a bid, we run a probit regression in which the dependent variable is whether a submitted bid wins a sealed-bid auction and the regressor of interest is an indicator for being a new bidder. We include number of bidder fixed effects and the auction covariates as control variables. In Table 4, column (2), we find that new bidders are more likely to win compared to existing bidders on average, conditional on bidding. Interpreting the magnitude of the probit coefficient, new bidders' estimated probability of winning is 0.06 higher than that of existing bidders at mean values of the auction covariates, at 0.23 relative to 0.17.

In Table 4, column (3), we go beyond the binary outcome of winning or losing and consider the rank within auction of each submitted bid. We define the normalized bid rank of a bid in a sealed-bid auction as $r \equiv \frac{n}{N+1}$ for the nth highest out of N bids, so $r \in (0, 1)$ for all $N \geq 1$. Despite the column (2) result that new bidders are more likely to win, column (3) finds no significant difference in average bid rank between new and existing bidders. Initially puzzling, these results can be explained upon inspecting the distribution of bid rank for each bidder group. We observe that, compared to existing bidders, the bid rank of new bidders has higher variance. As a result, differences in bid rank wash out on average, but a new bidder is more likely to submit the top bid. This speaks to the heterogeneity or non-monolithic nature of bidders being newly drawn to the online auctions.

3.3 Post-auction outcomes

The evidence suggests that after moving online, NMSLO auctions experienced an increase in auction revenue that is driven by new participants who increased competition substantially. As an additional consideration, we examine whether post-auction outcomes, including drilling and production, differ between tracts won by these new bidders and tracts won by existing bidders. New bidders, by definition, exist only in the online auction era, so we compare new and existing bidder outcomes using only the leases auctioned online. We define a tract as drilled post-auction if at least one well is spudded on the tract after its auction date. We define a tract's post-auction production as cumulative production from its post-auction well(s), in barrel-of-oil-equivalents (BOE). It should be noted that the vast majority of leased tracts are not drilled; in our data sample, the percentage of tracts drilled is around 5% overall. The low drilling rate is not unusual; Bhattacharya, Ordin and Roberts (2022) also observe that drilling was fairly rare for the leases auctioned during 1994–2012.

In Table 5, columns (1) and (2), we estimate a linear probability model and probit model of post-auction drilling, respectively. We do not detect a difference in drilling rates between tracts won by new versus existing bidders. Covariates that do seem most correlated with drilling, in order of smallest p-values, include the heatmap index, tract acreage, the annual rental rate, and the price of crude oil. Similarly, in Table 5, column (3), we check for a relationship between a new bidder winning the lease and realized post-auction production; again, we do not detect a relationship. Overall, we find no evidence in our data that post-auction outcomes are different for leases won by new versus existing bidders.

Table 5: Analysis of drilling and production for leases auctioned online

	(1) LPM: drilled	(2) probit: drilled	(3) OLS: log prod.
won by new bidder	-0.002	0.082	-0.117
	(0.019)	(0.201)	(0.210)
sealed bid auction	-0.024	-0.238	-0.312
	(0.019)	(0.188)	(0.209)
heatmap index	0.180	2.105	1.806
	(0.043)	(0.448)	(0.499)
drilled before	0.004	0.012	0.100
	(0.025)	(0.299)	(0.272)
cumulative BOE pre-auction (log)	0.002	0.013	-0.003
	(0.003)	(0.026)	(0.027)
primary rental=$1/acre	-0.044	-0.530	-0.297
	(0.026)	(0.273)	(0.280)
acres (in units of 100)	0.028	0.434	0.339
	(0.010)	(0.135)	(0.111)
WTI spot crude oil price (log)	0.216	2.303	1.286
	(0.126)	(1.253)	(1.460)
Henry Hub Natural Gas Spot Price (log)	-0.010	0.345	0.224
	(0.125)	(0.920)	(1.368)
Upstream Capital Cost Index	-0.005	-0.066	-0.051
	(0.009)	(0.079)	(0.111)
Upstream Operating Cost Index	-0.005	-0.008	0.016
	(0.008)	(0.060)	(0.087)
constant	-0.050	-9.496	-7.880
	(1.168)	(11.062)	(13.037)
Observations	605	605	605
R^2	0.085		0.075
Adjusted R^2	0.068		0.058
Pseudo R^2		0.214	

Notes: The regression sample includes all online auctions. The dependent variable in columns (1)–(2) is an indicator for at least one well being spudded on the auctioned tract after the auction date. The dependent variable in column (3) is the log of cumulative production from a tract's post-auction well(s), in barrel-of-oil-equivalents (BOE).

4 Conclusion

Government and public sector auctions are important and ubiquitous. This paper documents both the revenue impact of moving such an auction from offline to online implementation and the source of that effect. Our goal is to provide practitioners with detailed and direct evidence of the potential benefits of online auctions, even in a developed country context where auctioneer corruption is not a concern. We leverage data from oil and gas leases auctioned by the New Mexico State Land Office (NMSLO), the features of which facilitate an online versus offline comparison. Conditional on auction covariates, we find a large revenue difference in favor of online auctions, explained by a large increase in the number of bidders. Inspection of bidders' identities reveals that this increase is driven by an inflow of new participants who had not participated when the auctions were conducted offline. We do not detect a difference in post-auction outcomes between new and existing participants. In light of what we know from the auction literature, the magnitude of increased competition here suggests that moving auctions online deserves priority consideration, especially as the required innovation is relatively simple. For auction programs that are still run offline, online auctions would be well worth a trial.

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A Appendix

Table A1: Subsample analysis, Table 2-(1) and Table 3-(1)

	(1)	(2)	(3)	(4)
dependent variable:	log winning bid/acre		# bidders	
subsample:	top	non-top	top	non-top
online	1.357	0.434	3.047	1.430
	(0.448)	(0.135)	(0.797)	(0.337)
sealed bid auction	0.521	0.647		
	(0.226)	(0.083)		
heatmap index	1.585	1.092	0.790	1.339
	(0.475)	(0.179)	(0.966)	(0.428)
drilled before	-0.052	0.018	-0.152	-0.379
	(0.247)	(0.112)	(0.495)	(0.280)
cumulative BOE pre-auction (log)	0.017	0.014	0.128	0.030
	(0.026)	(0.011)	(0.046)	(0.026)
cumulative BOE post-auction (log)	0.089	0.057	0.180	-0.050
	(0.020)	(0.030)	(0.059)	(0.052)
primary rental=\$1/acre	0.000	0.100	0.000	0.460
	(.)	(0.125)	(.)	(0.267)
acres (in units of 100)	0.140	0.005	-0.248	0.103
	(0.113)	(0.046)	(0.215)	(0.127)
WTI spot crude oil price (log)	0.523	0.883	-1.137	0.192
	(0.841)	(0.369)	(1.307)	(0.855)
Henry Hub Natural Gas Spot Price (log)	-0.040	-0.348	-0.692	-0.213
	(0.877)	(0.323)	(1.257)	(0.749)
Upstream Capital Cost Index	0.060	-0.103	-0.248	-0.283
	(0.059)	(0.031)	(0.130)	(0.070)
Upstream Operating Cost Index	0.005	0.050	0.128	0.184
	(0.054)	(0.029)	(0.117)	(0.059)
constant	-15.935	6.034	24.878	13.557
	(8.551)	(3.902)	(17.249)	(8.279)
Sample: sealed (S) or ascending (O) auctions	S+O	S+O	S	S
Observations	210	630	144	328
R^2	0.318	0.260	0.238	0.222
Adjusted R^2	0.280	0.246	0.181	0.195

Notes: This table conducts subsample analyses of Table 2, column (1) and Table 3, column (1). The “top” subsample consists of leases in the top quartile of heatmap indices, and the “non-top” subsample consists of all other leases.

Table A2: Table 2 and Table 3-(1) analysis with post-double-selection method

	(1)	(2)	(3)	(4)	(5)	(6)
		log winning bid/acre			log bid/acre	# bidders
online	0.544	0.369	0.705	0.348	0.292	1.871
	(0.146)	(0.160)	(0.211)	(0.187)	(0.114)	(0.313)
sealed bid auction	0.619	0.383				
	(0.080)	(0.149)				
online × sealed bid auction		0.330				
		(0.176)				
WTI spot crude oil price (log)	0.697	-1.597	-2.480	7.363	-5.777	3.051
	(2.030)	(13.219)	(2.457)	(3.230)	(2.877)	(10.978)
Upstream Capital Cost Index	-0.033	-0.124				
	(0.032)	(0.297)				
heatmap index × heatmap index	0.159	0.143		0.538		
	(0.159)	(0.157)		(0.272)		
heatmap index					-5.501	
					(2.385)	
heatmap index × cumulative BOE post-auction (log)	-0.005	-0.003		0.021		
	(0.026)	(0.027)		(0.006)		
heatmap index × WTI spot crude oil price (log)	-0.076	0.038	0.547	-0.911	1.370	0.072
	(0.373)	(0.377)	(0.323)	(0.760)	(0.558)	(1.303)
heatmap index × Henry Hub Natural Gas Spot Price (log)	0.461	0.168		0.156		-0.964
	(0.579)	(0.573)		(1.028)		(1.146)
cumulative BOE post-auction (log) × WTI spot crude oil price (log)	0.031	0.030				
	(0.033)	(0.034)				
WTI spot crude oil price (log) × Upstream Operating Cost Index	0.003	0.098	0.003	-0.014	-0.005	-0.031
	(0.009)	(0.043)	(0.009)	(0.011)	(0.005)	(0.033)
Henry Hub Natural Gas Spot Price (log) × Upstream Operating Cost Index	-0.015	-0.333	-0.001	-0.007	-0.001	0.023
	(0.016)	(0.143)	(0.003)	(0.027)	(0.002)	(0.032)
WTI spot crude oil price (log) × Upstream Capital Cost Index		-0.078				
		(0.085)				
Henry Hub Natural Gas Spot Price (log) × Upstream Capital Cost Index		0.316				
		(0.140)				
heatmap index × Upstream Capital Cost Index			-0.002		0.007	
			(0.007)		(0.005)	
cumulative BOE post-auction (log) × primary rental=$1/acre				0.023		
				(0.043)		
cumulative BOE pre-auction (log) × primary rental=$1/acre					0.024	
					(0.011)	
cumulative BOE pre-auction (log) × Henry Hub Natural Gas Spot Price (log)					0.017	
					(0.008)	
heatmap index × Upstream Operating Cost Index						0.016
						(0.028)
constant	4.474	17.298	5.866	-8.868	26.696	-1.806
	(5.209)	(50.285)	(4.856)	(5.915)	(10.356)	(21.043)
Sample: sealed (S) or ascending (O) auctions	S+O	S+O	S	O	S	S
Observations: auction-level (a) or bid-level (b)	a	a	a	a	b	a
Observations	840	840	472	368	2002	472

Notes: This table repeats the analysis in Table 2 but uses the post-double-selection method of Belloni, Chernozhukov and Hansen (2014), allowing second-order terms and interactions of the control variables in Table 2. The method was implemented using the `pdslasso` program from Ahrens, Hansen and Schaffer (2018).